MOCK EXAMS
SCIENCE
2 & 1
Essay and Objectives
2 hours 10 minutes 2 & 1

Name:.....

School: .....

Index Number:.....

Date: .....

Signature: .....

Mock Examination

SCIENCE 2 & 1 2 hours

Do **not** *open this booklet until you are told to do so. While you are waiting, read and observe the following instructions carefully. Write your* **name** *and* **index number** *in the spaces provided above.*

This booklet consist of two papers. **Paper 1** *and* **Paper 2** *both of which must be taken. The two papers will be a composite paper to be taken at aa siting.*

1. **PAPER 1 (OBJECTIVE)**
   This will be objective paper made up of **40 questions** drawn from the entire syllabus. All the questions must be answered in **45 minutes**.

2. **PAPER 2 (ESSAY)**
   This shall be an essay paper made of **2** sections. Section **A** and **B** to be answered in **1 hour 25 minutes.**

   Section A: will be **one** compulsory question consisting of **four** sub questions testing in practical skills specified in Agriculture, Biology, Chemistry and Physics aspects of the examinations syllabus. This will attract **40 marks**.

   Section B: will consist of **four** questions. **Each** question shall be blend of Agriculture, Biology, Chemistry and Physics aspects of the examinations syllabus. Candidates will be required to answer any **three** of the question for **60 marks.**

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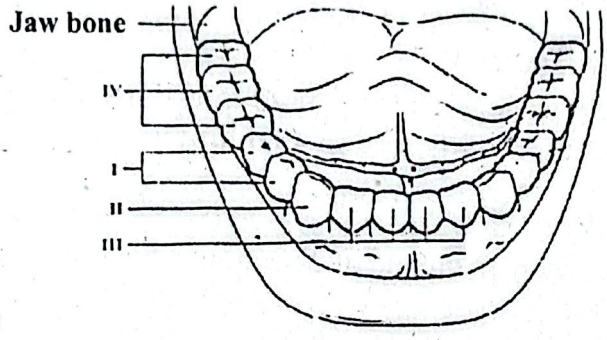
PAPER 2 1 hour 15 minutes
ESSAY
[100 marks]

his paper is an essay paper made of **2 sections**. Section **A** and **B** to be answered in **hour 15 minutes.** Answer Question **1** in Section I and any other **three** question in ection **II.**
nswer all the questions in your answer booklet.
redit will be given for clarity of expression and orderly presentation of material.

PART I
[40 marks]

Answer all of Question 1

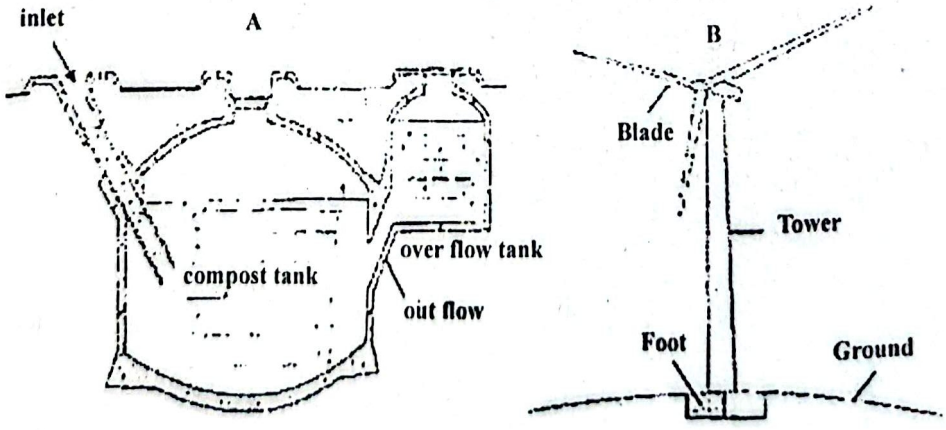
1. (a) The diagram below shows the arrangement of the different types of teeth in the lower jaw of humans. *Study the figure carefully and answer the questions that follow*



(i) Identify the parts labelled **I, II, III** and **IV**
(ii) State structural adaptations of **I, II** and **III**
(iii) Identify **one** common disease that affects the teeth
(iv) Give any **two** causes of the disease identified in a (iii) above

[10 marks]

(b) The diagrams below are illustrations of devices used to generate energy from various sources. *Study them carefully and answer the questions that follow*:



(i) Identify the devices labelled as **A** and **B**

(ii) Explain the type of energy source generated by both devices shown above

(iii) Give **two** advantages of the energy source explained in b (ii) above

(iv) State **one** major use of the devices

α) A

β) B

(v) State the energy transformations involved in the operation of the device **B**

vi Mention **one** device that can be used to harness energy from the sun [10 marks]

(c) The diagram below is a table of atoms of various elements with their respective atomic and mass numbers. *Study the table carefully and answer the questions that follow.*

Atom	Atomic number	Mass number
V	3	7
W	18	39
X	15	33
Y	10	20
Z	9	19

(i) How many neutrons are there in atom **X**?

(ii) Find the number of electrons in atom **Z**

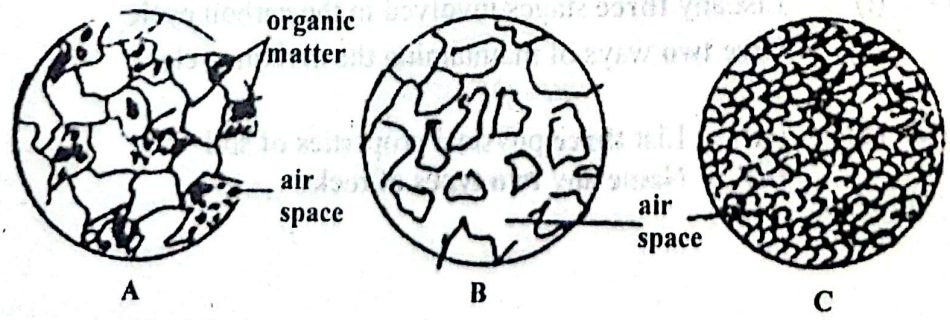
(iii) Which of the atom(s) are noble gases? Explain your answer

(iv) Explain how the atom **V** becomes an ion

(v) Identify any **two** atoms that will form negative ions

[10marks]

(d) The diagrams below illustrate different types of soil. *Study them carefully and answer the questions that follow*:



(i) Identify **each** of the soil types labelled **A, B** and **C.**
(ii) Give **one** reason for **each** of your answers in d (i) above
(iii) If it rains continuously for three hours, which of the soil types
(α) will dry faster?
(β) will easily become waterlogged?
(iv) State **two** reasons why soil type **B** is not suitable for crop production
[10 marks]

SECTION B
[60 marks]
*Answer **three** questions **only** in this section.*

2. (a) (i) Mention any **two** farming systems that involve raising animals.
(ii) Give ***three*** benefits derived from raising animals [5marks]

(b) (i) Name any **two** developmental stages of the mosquito found in water
(ii) Mention **two** activities to reduce the population of mosquito
[4 marks]

(c) A table tennis ball of mass 2.5 kg is dropped from a tower of height 7 m.
(i) State the type of energy the ball possessed on top of the tower.
(ii) Name the **two** forms of energy possessed by the ball as it falls
(iii) Calculate the value of the energy in c (i) [Take $g = 10m/s^2$]
[6marks]

(d) (i) Give **two** differences between
(α) atoms and ions
(β) protons and electrons
(ii) Why do elements in the same group have similar chemical properties [5marks]

3. (a) (i) List **three** sources of energy considered as non-renewable
(ii) Name **two** forms of energy used in our homes [5marks]

(b) (i) Draw the structure of aluminium atom and write down its electronic configuration
(ii) From the structure above, identify the group and period of aluminium
[5marks]

(c) (i) List any **three** stages involved in the carbon cycle
(ii) Give **two** ways of maintaining the carbon cycle [5marks]

(d) (i) List **three** physical properties of soil
(ii) Name any **two types of rocks** [5marks]

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4. (a) (i) List **three** examples of *monogastrics*.
(ii) State **two** differences between ***ruminants*** and ***non-ruminants***
[5marks]

(b) (i) Give **two** characteristics of liquids
(ii) Name **two** processes that change matter from one phase to another
(iii) Draw diagrams to show the arrangement of molecules in solids, liquids and gases [7marks]

(c) (i) Define ***kinetic energy***
(ii) A 100 kg toy car moves with a velocity of 2 m/s on a smooth horizontal ground. Calculate its kinetic energy [4marks]

(d) (i) What is ***a food web***?
(ii) Construct a food chain using any four of the following organisms:
grasshopper, goat, lion, grass, lizard, snake, hawk, man, cassava leaf
[4marks]

5. (a) List the **three** types of seed bed [3marks]

(b) (i) Give **two** characteristics each of the following planets
α) Venus
β) Earth
ii) Give **two** reasons why there is no life on planet Jupiter
(iii) Name **one** other example of outer planet [7marks]

(c) Name the cell organelle responsible for the functions below
(i) Controls cell division
(ii) Storage of dissolved minerals and chemicals
(iii) Site for cellular respiration
(iv) Synthesis of protein
(v) Site for photosynthesis in green plants [5marks]

(d) (i) Differentiate between ***solution*** and ***suspension***
(ii) Name one method used to separate the components of the following mixtures
(α) Iron filings and sand
(β) Alcohol and water
(γ) Rice particles in water [5marks]

End of paper

Turn Over

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Answer all the questions

Each question is followed by **four** *options lettered A to D. Find the correct option for* **each** *question and shade* **in pencil** *on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only* **one** *answer to* **each** *question. An examples is given below:*

Which of the following substances is not an element?
A. Oxygen
B. Ammonia
C. Sodium
D. Aluminium

The correct answer is* Ammonia, *which is lettered B and therefore answer spaces B would be shaded.

[A] [B] [C] [D]

Think carefully before you shade the answer spaces; erase completely any any answers you wish to change.
Do all rough work on this question paper
Now answer the following questions

1. The easiest method of sowing seeds on the field is
   A. band placement
   B. broadcasting method
   C. ring method
   D. drilling method

2. Which of the following is found in the nucleus of an atom?
   I. Protons II. Electrons
   III. Neutrons

   A. I and II only
   B. II and III only
   C. I and III only
   D. I, II and III

3. Light rays bend when they move from one medium to another of different density. This is due to light
   A. reflection
   B. refraction
   C. absorption
   D. dispersion

4. Which of the following components of soil absorb heat from the sun to regulate soil temperature due to its dark colour?
   A. Humus B. Water
   C. Air D. Soil particles

5. The main reason for including groundnut in crop rotation programme
   A. to improve drainage
   B. improve aeration
   C. control weed growth
   D. fix nitrogen into the soil

6. In a food chain, the organism that receives the lowest energy is the
   A. producer
   B. secondary consumer
   C. primary consumer
   D. tertiary consumer

7. Which of the following substances has a definite shape?
   A. Air B. Stone
   C. Kerosene D. Water

8. Which of the following cell organelles is responsible for cell division?
   A. Mitochondrion
   B. Nucleus
   C. Kerosene
   D. Chloroplast

9. Which of the following methods is used to obtain water from salt solution?
   A. Filtration
   B. Evaporation
   C. Distillation
   D. Sedimentation

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10. Mass number of an atom is
A. the sum of neutrons and protons
B. the sum of protons and electrons
C. always more than the atomic number
D. always less than the atomic number

11. An electronic component which stores electric charges is
A. capacitor B. transistor
C. diode D. inductor

12. In carbon cycle, plants release carbon dioxide to the atmosphere through
A. respiration
B. decomposition
C. transpiration
D. photosynthesis

13. Which of the following physical properties of soil is determined by feel?
A. Texture B. Drainage
C. Structure D. Capillarity

14. When an atom loses an electron, it forms
A. a molecule B. a compound
C. a cation D. an anion

15. An example of monogastric is
A. cattle B. sheep
C. goat D. rabbit

16. Which of the following pairs of energy sources is environmentally friendly?
A. Coal and thermal energy
B. Wind and biogas
C. Wind and coal
D. Biogas and fossil fuel

17. The form of energy produced due to vibrations of objects is
A. light B. electricity
C. sound D. heat

18. Fine soil particles which stick together when wet are classified as
A. clay B. silt
C. sand D. loam

19. Electrons found in the outermost shell of an atom are called.....electrons
A. core B. valence
C. mobile D. lone pai

20. The farming system in which different crops are planted on the same piece of land at the same time is known as
A. crop rotation
B. mixed cropping
C. mixed farming
D. monoculture

21. A ball lying on the floor will continu in that state unless it is kicked. Which Newton's law of motion is demonstrated?
A. First law
B. Second law
C. Third law
D. Fourth law

22. A mango of mass 2 kg hangs 15 m o top of a tree. Determine its kinetic energy just before it hits the ground
A. 30.0 J B. 300.0
C. 225.0 J D. 3.0 J

23. Heat energy from the sun reaches th earth by
A. conduction B. reflection
C. convection D. radiation

24. Which of the following is **not** a deficiency disease?
A. Cholera B. Beriberi
C. Scurvy D. Goitre

25. The planet known as the red planet
A. Venus B. Mars
C. Jupiter D. Satur

Turn O

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Which of the following forms of energy is released during friction of two bodies?
A. Chemical energy
B. Mechanical energy
C. Nuclear energy
D. Thermal energy

Which of the following is **not** an abiotic factor?
A. Temperature
B. Producers
C. Humidity
D. Sunlight

The energy transformation that takes place when a torch light is switched on is
A. chemical energy to light energy to heat energy
B. electrical energy to chemical energy to heat energy
C. chemical energy to electrical energy to light energy
D. electrical energy to kinetic energy to light energy

). The type of seed bed that is made to lie below the surrounding ground is
A. spiral bed
B. sunken bed
C. flat bed
D. raised bed

). An atom has 20 nucleons and 9 protons. What is its neutron number?
A. 9 B. 10
C. 11 D. 12

1. The soil type which has a gritty feel when wet is
A. sand B. clay
C. silt D. gravel

2. The elements in the periodic table are arranged according to the
A. atomic number
B. atomic mass
C. mass number
D. neutron number

33. Chemical energy can be obtained from the following sources except
A. electricity B. charcoal
C. dry cell D. kerosene

34. The nearest planet to the sun is
A. Earth B. Mercury
C. Mars D. Venus

35. A liquid-liquid mixture can be separated by
A. filtration B. distillation
C. sublimation D. evaporation

36. The end products of digestion of protein is
A. amino acid B. glycerol
C. glucose D. fatty acid

37. Which of the following parts of the tooth is embedded in the jaw bone?
A. Gum B. Neck
C. Root D. Crown

38. Which of the following is an example of an inert gas?
A. Hydrogen B. Nitrogen
C. Chlorine D. Argon

39. The relative proportions of the different particle sizes in a soil sample is
A. soil temperature
B. soil structure
C. soil texture
D. soil profile

40. A substance made up of the same kind of atoms is called
A. a compound
B. a mixture
C. an element
D. an ion

END OF PAPER

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